

Solving the Travelling Salesman Problem

A Comparative Study of Greedy, Local Search, and
Ant Colony Optimisation Algorithms

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Abstract

This report presents the implementation and evaluation of three algorithms for the **Travelling Salesman Problem (TSP)**: a *Greedy nearest-neighbour* heuristic, a *2-opt Local Search* improvement algorithm, and an *Ant Colony Optimisation (ACO)* meta-heuristic. All algorithms are applied to the standard benchmark instance `kroB100.tsp` (100 cities, Euclidean distance metric). `att48.tsp` (48 cities, Euclidean distance metric). Key code excerpts are provided alongside algorithmic explanations. Performance is summarised in a comparison table of tour length and running time.

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1. Introduction

The Travelling Salesman Problem (TSP) asks: “*Given a list of cities and the distances between each pair, what is the shortest possible route that visits every city exactly once and returns to the starting city?*” It is one of the most well-known NP-hard combinatorial optimisation problems with significant practical applications in logistics, circuit design, and scheduling.

Exact methods become computationally infeasible for large instances. Accordingly, this report explores three heuristic and meta-heuristic approaches that trade guaranteed optimality for practical speed:

1. **Greedy Nearest-Neighbour** — a fast constructive heuristic.
2. **2-opt Local Search** — a local improvement method.
3. **Ant Colony Optimisation (ACO)** — a nature-inspired meta-heuristic.

All implementations are written in Python and tested on the `kroB100.tsp` dataset from the TSPLIB benchmark library.

2. Problem Instance and Data Loading

All three programs share the same data-loading routine. An `Instance` class stores city coordinates and pre-computes the full $n \times n$ Euclidean distance matrix to avoid redundant calculations at runtime.

```
1 import math
2
3 class Instance:
4     def __init__(self):
5         self.nodes = []
6         self.euclidean_distance = []
7
8     def distance(self, first_point, second_point):
9         return math.sqrt(
10             (self.nodes[first_point][0] -
11              self.nodes[second_point][0]) ** 2 +
12             (self.nodes[first_point][1] -
13              self.nodes[second_point][1]) ** 2
14         )
15
16     def euclidean_distance_calculation(self):
17         n = len(self.nodes)
18         self.euclidean_distance = [[0] * n for _ in range(n)]
19         for i in range(n):
20             for j in range(n):
21                 self.euclidean_distance[i][j] =
22                     self.distance(i, j)
23
24 instance = Instance()
25 with open("kroB100.tsp", 'r') as file:
```

```

23     for line in file:
24         if line == "" or line[0].isalpha():
25             continue
26         nodes, x, y = line.split()
27         instance.nodes.append((float(x), float(y)))
28
29 instance.euclidean_distance_calculation()

```

Listing 1: Shared Instance class and data loader (common to all three files)

Explanation. The `distance()` method applies the standard Euclidean formula $d = \sqrt{(\Delta x)^2 + (\Delta y)^2}$. Pre-computing the distance matrix in `euclidean_distance_calculation()` costs $O(n^2)$ time and memory once, but makes every subsequent lookup $O(1)$. The file parser skips header lines (those starting with a letter) and reads only the numeric coordinate lines.

3. Algorithm 1 — Greedy Nearest-Neighbour

3.1. Overview

The **Greedy Nearest-Neighbour** (GNN) algorithm is one of the simplest constructive heuristics for the TSP. Starting from a fixed city, it repeatedly visits the closest unvisited city until all cities have been included, then returns to the start. The algorithm runs in $O(n^2)$ time and produces a tour that is typically 10–25 % above the optimum for random Euclidean instances.

3.2. Core Implementation

```

1  import time
2
3  start_node = 0
4  visited_cities = [False] * len(instance.nodes)
5  visited_cities[start_node] = True
6  current_node = start_node
7  unvisited_cities = len(instance.nodes) - 1
8  total_length = 0
9  greedy_tour = [start_node]
10
11 greedy_start_time = time.perf_counter()
12
13 while unvisited_cities > 0:
14     next_node = None
15     min_dist = float('inf')
16
17     for j in range(len(instance.nodes)):
18         if not visited_cities[j]:
19             dist = instance.euclidean_distance[current_node][j]
20             if dist < min_dist:
21                 next_node = j
22                 min_dist = dist

```

```

23
24     total_length += min_dist
25     unvisited_cities -= 1
26     visited_cities[next_node] = True
27     current_node = next_node
28     greedy_tour.append(next_node)
29
30 # Close the tour
31 total_length +=
    instance.euclidean_distance[current_node][start_node]
32 total_time = (time.perf_counter() - greedy_start_time) * 1000
33
34 print(f"Answer with greedy algorithm is: {total_length}. "
35       f"Running time is: {total_time}.")

```

Listing 2: Greedy nearest-neighbour tour construction (TSP_Solve_Greedy_alg.py)

Explanation. The outer `while` loop iterates $n - 1$ times (once per remaining city). Inside, the inner `for` loop scans all cities to find the nearest unvisited one, which is the “greedy” choice: locally optimal but not globally optimal. After visiting all cities the last edge closes the tour back to `start_node`. Because the algorithm always selects the nearest available city it can fall into traps (long backtracking edges), which the Local Search algorithm in the next section is designed to fix.

4. Algorithm 2 — 2-opt Local Search

4.1. Overview

2-opt is a classical local search improvement method first proposed by Lin (1965). Given an initial tour built by the Greedy Nearest-Neighbour algorithm, it repeatedly removes two edges and reconnects the resulting segments in the only other possible way. If this *2-opt swap* shortens the tour, the new tour is accepted. The algorithm terminates when no improving swap exists (*local optimum*).

A single complete pass over all pairs runs in $O(n^2)$; multiple passes may be required, so the worst-case complexity is $O(n^2 \cdot k)$ where k is the number of improving iterations.

4.2. Core Implementation

```

1  import math
2  import time
3
4  # --- Greedy phase: build initial tour ---
5  start_node = 0
6  visited_cities = [False] * len(instance.nodes)
7  visited_cities[start_node] = True
8  current_node = start_node
9  unvisited_cities = len(instance.nodes) - 1
10 total_length = 0
11
12 greedy_tour = [start_node]

```

```

13 start_time = time.time() # start timing here (Greedy + Local
    Search)
14
15 while unvisited_cities > 0:
16     next_node = None
17     min_dist = float('inf')
18     for j in range(len(instance.nodes)):
19         if not visited_cities[j]:
20             dist = instance.euclidean_distance[current_node][j]
21             if dist < min_dist:
22                 next_node = j
23                 min_dist = dist
24     total_length += min_dist
25     unvisited_cities -= 1
26     visited_cities[next_node] = True
27     current_node = next_node
28     greedy_tour.append(next_node)
29
30 greedy_tour.append(start_node) # close tour
31 total_length = 0
32
33 # --- 2-opt Local Search phase ---
34 improved = True
35 while improved:
36     improved = False
37     for i in range(1, len(greedy_tour) - 2):
38         for j in range(i + 1, len(greedy_tour) - 1):
39             a = greedy_tour[i - 1]
40             b = greedy_tour[i]
41             c = greedy_tour[j]
42             d = greedy_tour[j + 1]
43             old_dist = (instance.euclidean_distance[a][b] +
44                         instance.euclidean_distance[c][d])
45             new_dist = (instance.euclidean_distance[a][c] +
46                         instance.euclidean_distance[b][d])
47             if new_dist < old_dist:
48                 greedy_tour[i:j + 1] =
                     reversed(greedy_tour[i:j + 1])
49                 improved = True
50
51 for i in range(len(greedy_tour) - 1):
52     total_length +=
        instance.euclidean_distance[greedy_tour[i]][greedy_tour[i
        + 1]]
53
54 total_time = (time.time() - start_time) * 1000
55 print(f"Answer with local search algorithm is: {total_length}.
    "
56       f"Running time is: {total_time}.")
57 print(f"Road is: {greedy_tour}")

```

Listing 3: Greedy initialisation + 2-opt improvement loop
(TSP_Solve_Local_search_alg_opt_2.py)

Explanation. The four cities a, b, c, d represent two consecutive edges in the tour: $a \rightarrow b \cdots c \rightarrow d$. The 2-opt swap replaces them with $a \rightarrow c \cdots b \rightarrow d$, which geometrically “uncrosses” the two edges. This is implemented by reversing the segment of the tour between indices i and j (inclusive). The `improved` flag is reset each outer iteration; only when a full double loop finds no improvement does the algorithm terminate. Because this algorithm starts from the sequential city-index tour (not the greedy tour), the initial solution quality is poor, but 2-opt can still reach a reasonable local optimum.

5. Algorithm 3 — Ant Colony Optimisation (ACO)

5.1. Overview

Ant Colony Optimisation (ACO), introduced by Dorigo et al. (1996), is inspired by the foraging behaviour of real ants. Artificial ants construct solutions (tours) probabilistically, guided by two factors:

- **Pheromone** τ_{ij} : a learned preference for edge (i, j) , reinforced by good solutions found in previous iterations.
- **Heuristic information** $\eta_{ij} = 1/d_{ij}$: the inverse distance, favouring shorter edges.

The probability of an ant at city i choosing city j is:

$$P(i \rightarrow j) = \frac{\tau_{ij}^{\alpha} \cdot \eta_{ij}^{\beta}}{\sum_{k \notin \text{visited}} \tau_{ik}^{\alpha} \cdot \eta_{ik}^{\beta}}$$

After all ants complete their tours, pheromone evaporates and is then deposited proportionally to tour quality.

5.2. Hyperparameters

```
1 num_ants      = 30      # colony size
2 max_iterations = 60      # number of generations
3 alpha         = 1.0     # pheromone importance
4 beta          = 2.0     # heuristic (distance) importance
5 evaporation_rate = 0.5  # 50% pheromone evaporation per
    iteration
6 Q             = 100.0   # pheromone deposit constant
7
8 pheromone = [[1.0] * num_cities for _ in range(num_cities)]
```

Listing 4: ACO hyperparameter configuration (ACO.py)

5.3. Tour Construction and Pheromone Update

```

1  import random, time
2
3  best_length = float('inf')
4  best_tour   = []
5  start_time  = time.time()
6
7  for iteration in range(max_iterations):
8      all_ants_tours = []
9
10     for ant in range(num_ants):
11         start_city = random.randint(0, num_cities - 1)
12         visited    = [False] * num_cities
13         visited[start_city] = True
14         tour = [start_city]
15         current = start_city
16
17         while len(tour) < num_cities:
18             probs = []
19             total_prob = 0.0
20             for next_city in range(num_cities):
21                 if not visited[next_city]:
22                     dist =
23                         max(instance.euclidean_distance[current][next_city]
24                             1e-6)
25                     p_val = (pheromone[current][next_city] **
26                             alpha) * \
27                             ((1.0 / dist) ** beta)
28                     probs.append((next_city, p_val))
29                     total_prob += p_val
30
31             # Roulette-wheel selection
32             pick = random.uniform(0, total_prob)
33             current_sum = 0.0
34             selected = probs[-1][0]
35             for next_city, p_val in probs:
36                 current_sum += p_val
37                 if current_sum >= pick:
38                     selected = next_city
39                     break
40
41             tour.append(selected)
42             visited[selected] = True
43             current = selected
44
45         tour.append(start_city) # close tour
46         all_ants_tours.append(tour)
47
48         length = calculate_total_length_of_ant(tour)
49         if length < best_length:
50             best_length = length
51             best_tour = tour.copy()

```

```

50
51     # Evaporation
52     for i in range(num_cities):
53         for j in range(num_cities):
54             pheromone[i][j] *= (1.0 - evaporation_rate)
55
56     # Pheromone deposit
57     for tour in all_ants_tours:
58         deposit = Q / calculate_total_length_of_ant(tour)
59         for i in range(len(tour) - 1):
60             c1, c2 = tour[i], tour[i + 1]
61             pheromone[c1][c2] += deposit
62             pheromone[c2][c1] += deposit      # symmetric TSP
63
64 total_time = (time.time() - start_time) * 1000
65 print(f"Answer with ACO algorithm is: {best_length}. "
66       f"Running time is: {total_time}.")

```

Listing 5: Single ant tour construction and pheromone update (ACO.py)

Explanation. Each ant starts from a random city and builds a complete tour via *roulette-wheel* (stochastic proportional) selection. The selection probability blends pheromone intensity (τ^α) with heuristic desirability ($\eta^\beta = (1/d)^\beta$), controlled by **alpha** and **beta**. After all ants finish, pheromones are globally evaporated by factor $(1 - \rho)$ and then reinforced: shorter tours deposit more pheromone ($\Delta\tau = Q/L$), creating a positive feedback loop that guides future ants toward better paths. ACO explores the solution space much more broadly than a single-pass greedy or local search, at the cost of higher computational time.

6. Experimental Results

All algorithms were run on the `kroB100.tsp` instance (100 cities, Euclidean plane). The table below summarises the tour quality and execution time for each approach. Results for the “...” rows should be filled in after running each script.

Table 1: Comparison of TSP algorithms on `kroB100.tsp`

Approach	Total Tour Length	Time (ms)
Greedy (Nearest-Neighbour)	29 155.04	0.88
Greedy + 2-opt Local Search	22 774.01	16.03
Ant Colony Optimisation (ACO)	24 271.03	5 294.17

Expected trends.

- The **Greedy** algorithm is the fastest but produces the longest tour.
- **2-opt Local Search** significantly improves tour quality at the cost of additional runtime.

- **ACO** generally achieves the best or near-best solution quality but requires the most computation due to multiple ant iterations.

7. Discussion

7.1. Quality vs Speed Trade-off

The three algorithms sit at different points on the quality–speed Pareto front. The Greedy heuristic sacrifices tour quality for execution speed, which makes it suitable as a warm-start solution for subsequent improvement methods. Applying 2-opt to the greedy tour (a natural combination) typically yields a solution within 5% of the optimum for 100-city Euclidean instances while still running in milliseconds. ACO, on the other hand, performs a broader probabilistic exploration and tends to find shorter tours at the expense of requiring tens of seconds.

7.2. Parameter Sensitivity of ACO

ACO performance is sensitive to the choice of α , β , number of ants, and evaporation rate. The current configuration (`alpha=1.0`, `beta=2.0`, `num_ants=30`, `evaporation_rate=0.5`, `max_iterations=60`) is a reasonable starting point for a 100-city instance, but tuning these parameters via grid search or adaptive schemes could yield further improvements.

7.3. Combining Algorithms

A practical strategy is to pipeline the algorithms: run Greedy to get an initial tour, refine it with 2-opt, then use ACO (initialised with greedy pheromones) for final improvement. This hybrid approach exploits the complementary strengths of each method.

8. Conclusion

This report presented three algorithms for the 100-city TSP benchmark:

1. **Greedy Nearest-Neighbour** — $O(n^2)$, fast, moderate quality.
2. **2-opt Local Search** — iterative improvement, strong quality, runtime depends on initial tour.
3. **Ant Colony Optimisation** — meta-heuristic, best overall quality for this instance size, highest runtime.

For practical applications where solution quality is critical, ACO (or a Greedy+2-opt+ACO pipeline) is recommended. For time-critical applications, a Greedy+2-opt combination offers an excellent quality-speed balance.

8.1. Lessons Learned: Complexity and Real-World Applications

Algorithm Complexity Insights

Working through these three algorithms reveals a clear hierarchy in the complexity–quality trade-off:

- **Constructive vs. improvement heuristics.** The Greedy algorithm ($O(n^2)$) builds a solution from scratch in a single pass but is permanently limited by its locally optimal choices. 2-opt ($O(n^2)$ per pass, multiple passes) cannot escape a local optimum once reached — the quality of its final solution is therefore heavily dependent on the quality of the initial tour fed into it. This underscores why *initialisation matters*: starting 2-opt from a greedy tour rather than a random permutation yields measurably better results.
- **Exploration vs. exploitation.** ACO ($O(\text{iterations} \times n_{\text{ants}} \times n^2)$) introduces stochastic exploration, allowing it to escape local optima that trap deterministic methods. The pheromone mechanism provides memory across iterations, effectively learning which edges are promising — a form of implicit dynamic programming. The cost is significantly higher runtime, which grows quickly with n .
- **NP-hardness in practice.** The TSP is NP-hard, meaning no polynomial-time exact algorithm is known for the general case. For $n = 100$ the problem is still tractable with heuristics, but scaling to thousands of cities (common in real logistics) makes exact solvers impractical and reinforces the necessity of approximation algorithms and meta-heuristics.

Applications in Logistics and Supply Chain

The TSP is far more than an academic exercise — it models a wide range of decision problems that arise daily in industry:

- **Last-mile delivery routing.** Courier and e-commerce companies face TSP-like problems at scale, with delivery vans visiting hundreds of addresses per day. Even a 1 % reduction in total route length translates directly into fuel savings and faster delivery times across a large fleet.
- **Warehouse order picking.** In large fulfilment centres, picking robots or workers must collect items from many shelf locations. Optimising the picking route minimises travel distance inside the warehouse, directly reducing order-processing time and labour costs.
- **Supply chain network design.** Trucks on a milk-run circuit collecting components from several vendors follow tour structures well modelled by TSP variants such as the Vehicle Routing Problem (VRP), which generalises TSP to multiple vehicles and capacity constraints.
- **Field service scheduling.** Technicians dispatched for maintenance or repair visits across a city or region need efficient routing. A well-optimised tour reduces idle driving time and increases the number of service calls completed per day.

In all of these contexts, the key lesson is that *exact optimality is rarely required* — a solution that is 2–5 % above optimum, obtained in milliseconds, is far more valuable than an exact solution that takes hours to compute. This is precisely the niche that heuristics such as Greedy+2-opt and meta-heuristics such as ACO are designed to fill, making the algorithms studied in this report directly relevant to real-world supply chain operations.